Energy Usage

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Abstract

Data set "Appliances Energy Prediction" from UC Irvine Machine Learning Repository was used in this study. The purpose of this study is create a predictive visual model and forecast of energy usage. We will focus on using the Time series Regression and Forecasting to analyze the data. We were able to separate training and test data to test the CI accuracy.

Introduction

"Appliances Energy Prediction" is data set that includes house temperature and humidity conditions every 3.3 min then averaged the 3 for data every 10 min for 4 1/2 months. We will use this data to create a predictive visual model and forecast of energy usage based on these 4 1/2 months. The data set includes 29 variables ranging from with humidity and temperature readings from 9 different locations of the house including: bathroom, kitchen, bedroom to living room. We will mainly focus Appliances and Date variables. We will use Exploratory Data Analysis, time series regression with linear model regression, and forecast to predict Confidence Interval energy usage. The model we created successfully which visualize energy usage over time and concluded there no is trend and no seasonal pattern. The data was downloaded from UCI Machine Learning Repository and analyzed using R studio.

Sections

Test Split and Time Series Regression

```r
library(dplyr)
library(readr)
library(forecast)
#DATA
energydata_complete <- read_csv("energydata_complete.csv")
energydata_complete <- energydata_complete %>%
  mutate(date = as.Date(date)) %>%
  group_by(date) %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE))

#Train and Test
train_size <- nrow(energydata_complete) - 10
train_data <- energydata_complete[1:train_size, ]
train_data <- train_data %>%
  mutate(lag_date = date - min(date))
test_data <- energydata_complete[(train_size + 1):nrow(energydata_complete), ]
```

```r
#Validation
dim(train_data)
```

```
## [1] 128  30
```

```r
dim(test_data)
```

```
## [1] 10 29
```

```r
range(train_data$date)
```

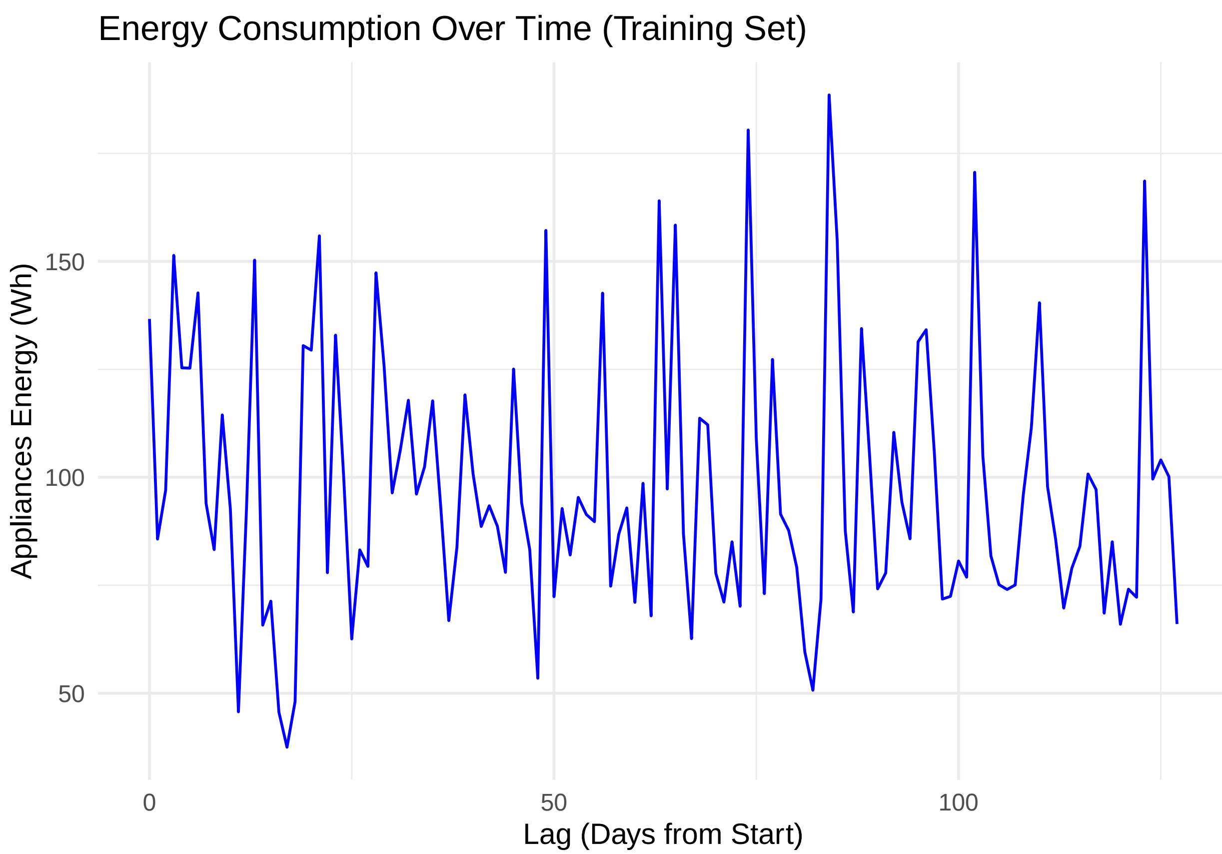
```
## [1] "2016-01-11" "2016-05-17"
```

```r
range(test_data$date)
```

```
## [1] "2016-05-18" "2016-05-27"
```

Training set Plot and analyze

```r
library(ggplot2)

# Plot
ggplot(train_data, aes(x = lag_date, y = Appliances)) +
  geom_line(color = "blue") +
  ggtitle("Energy Consumption Over Time (Training Set)") +
  xlab("Lag (Days from Start)") +
  ylab("Appliances Energy (Wh)") +
  theme_minimal()
```



i. There does not seem to be a trend of due inconsistent spikes and drops. ii. There is no seasonal like pattern as data covers 4 and 1/2 months of date each day counting as lag from 1/11/2016. iii. There are certain spikes of appliances energy on random lags throughout the plotted 128 lags.

Stationary

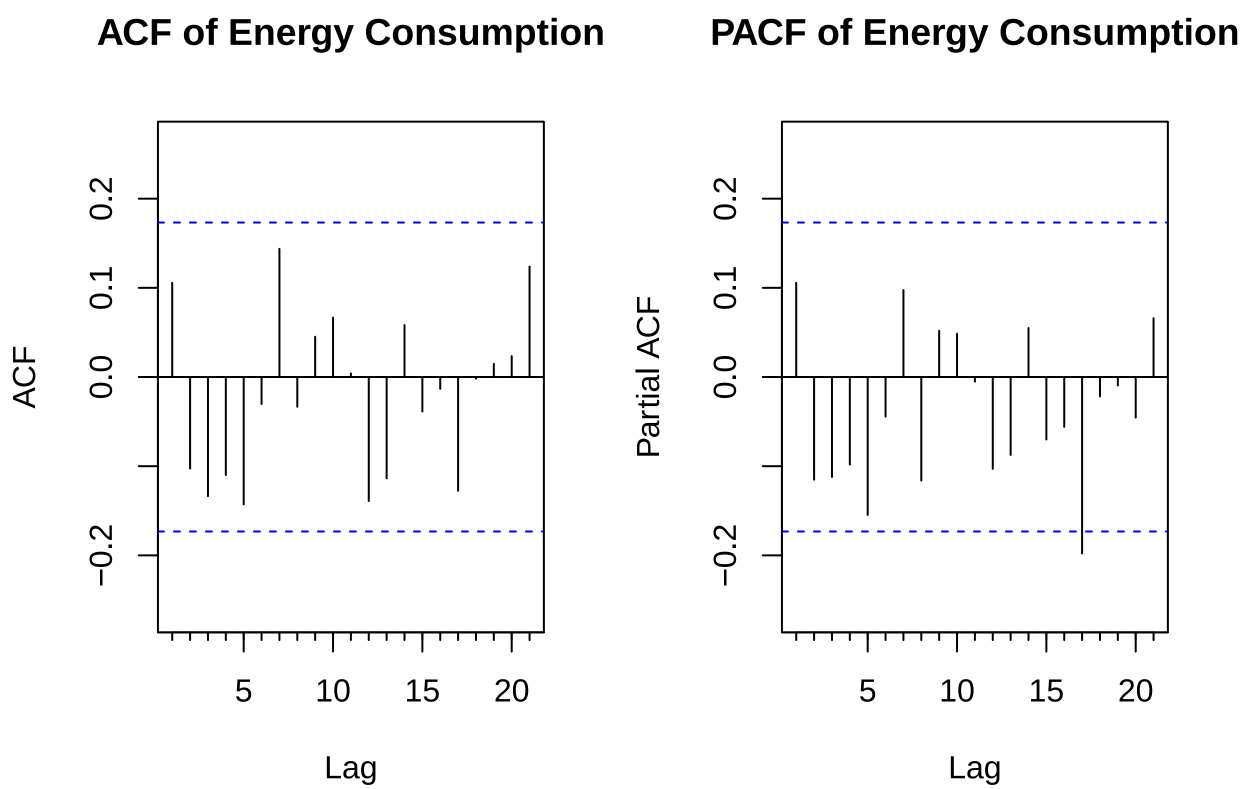
```r
library(tseries)

ts_train <- ts(train_data$Appliances, frequency = 1)
#ADF
adf_test <- adf.test(ts_train)
print(adf_test)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  ts_train
## Dickey-Fuller = -6.2992, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

Given that there are no trend, seasonal pattern and maintain constant variance and mean. Finally, p-value is less than 0.05 meaning we reject the null hypothesis and series is already stationary. Box Jenkins method is not needed for this data.

Plot ACF and PACF

```r
# Plot ACF and PACF
par(mfrow = c(1,2))
Acf(ts_train, main = "ACF of Energy Consumption")
Pacf(ts_train, main = "PACF of Energy Consumption")
```



Both ACF and PACF does not tail off or cut off at specific lags as no spikes are higher than the Confidence Interval. This determines p = 0 and q = 0 as all the spikes in all the lags are within the CI expect lag 17 in PACF however, since there are no previous lags we will ignore this significant spike and choose p and q as 0. Therefore, we will use Model ARIMA(0,0,0) and ARIMA(1,0,0) to test.

Fit Models

```r
# ARIMA(0,0,0)
arima <- Arima(train_data$Appliances, order = c(0, 0, 0))
summary(arima)
```

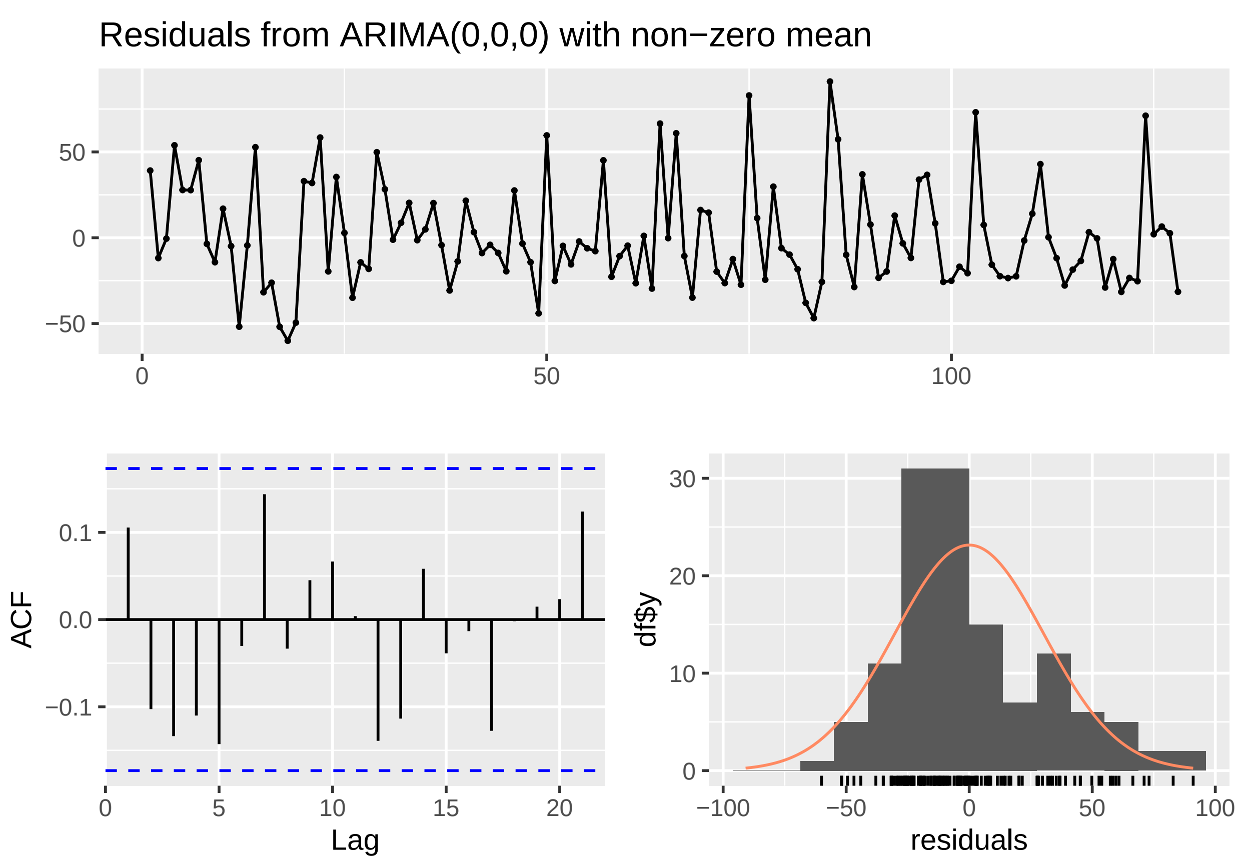
```
## Series: train_data$Appliances
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##          mean
```

```
##        97.5288
## s.e.    2.6670
##
## sigma^2 = 917.6:  log likelihood = -617.71
## AIC=1239.43   AICc=1239.52   BIC=1245.13
##
## Training set error measures:
##                        ME     RMSE      MAE       MPE     MAPE      MASE
## Training set 2.009504e-14 30.17313 23.52182 -9.671159 26.12336 0.7834752
##                   ACF1
## Training set 0.1055503
```

```
#  AIC
aic <- AIC(arima)
k <- length(coef(arima))
n <- length(train_data$Appliances)
aicc <- aic + (2 * k * (k + 1)) / (n - k - 1)
aicc
```

```
## [1] 1239.46
```

```
# Residuals
checkresiduals(arima)
```




```
##
```

```
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,0) with non-zero mean
## Q* = 13.648, df = 10, p-value = 0.1896
##
## Model df: 0.   Total lags used: 10
```

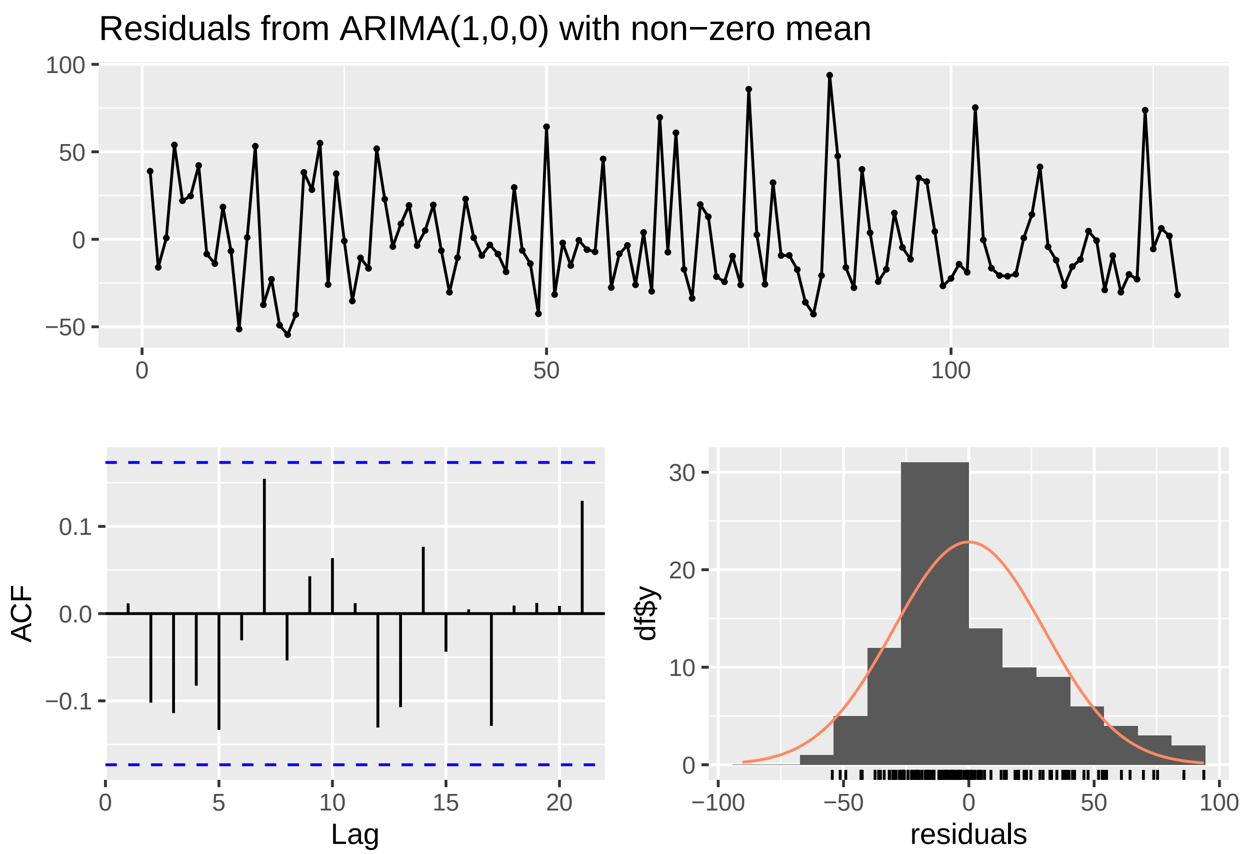
```r
# ARIMA(1,0,0)
arima1 <- Arima(train_data$Appliances, order = c(1, 0, 0))
summary(arima1)
```

```
## Series: train_data$Appliances
## ARIMA(1,0,0) with non-zero mean
##
## Coefficients:
##          ar1     mean
##       0.1070  97.5359
## s.e.  0.0885   2.9668
##
## sigma^2 = 914.3:  log likelihood = -616.99
## AIC=1239.97   AICc=1240.17   BIC=1248.53
##
## Training set error measures:
##                       ME     RMSE      MAE       MPE     MAPE      MASE
## Training set -0.03447116 30.00086 23.23186 -9.455758 25.57151 0.7738169
##                    ACF1
## Training set 0.01180097
```

```r
#  AIC
aic1 <- AIC(arima1)
k1 <- length(coef(arima1))
n1 <- length(train_data$Appliances)
aicc1 <- aic1 + (2 * k1 * (k1 + 1)) / (n1 - k1 - 1)
aicc1
```

```
## [1] 1240.07
```

```r
# Residuals
checkresiduals(arima1)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,0) with non-zero mean
## Q* = 11.078, df = 9, p-value = 0.2704
##
## Model df: 1.   Total lags used: 10
```

ARIMA(0,0,0): RMSE: 30.17313 MAE: 23.52182 MAPE: 26.12336 ACF1: 0.10555 ARIMA(0,0,0): AICc = 1239.46

ARIMA(1,0,0): RMSE: 30.00086 MAE: 23.23186 MAPE: 25.57151 ACF1: 0.01180 ARIMA(1,0,0): AICc = 1240.07

Yes we used AICc and BIC to choose "best model". ARIMA(0,0,0) has lower AICc and BIC value meaning it is slightly better fit. We can also see lower values in ACF for ARIMA(0,0,0) compared to ARIMA(1,0,0).

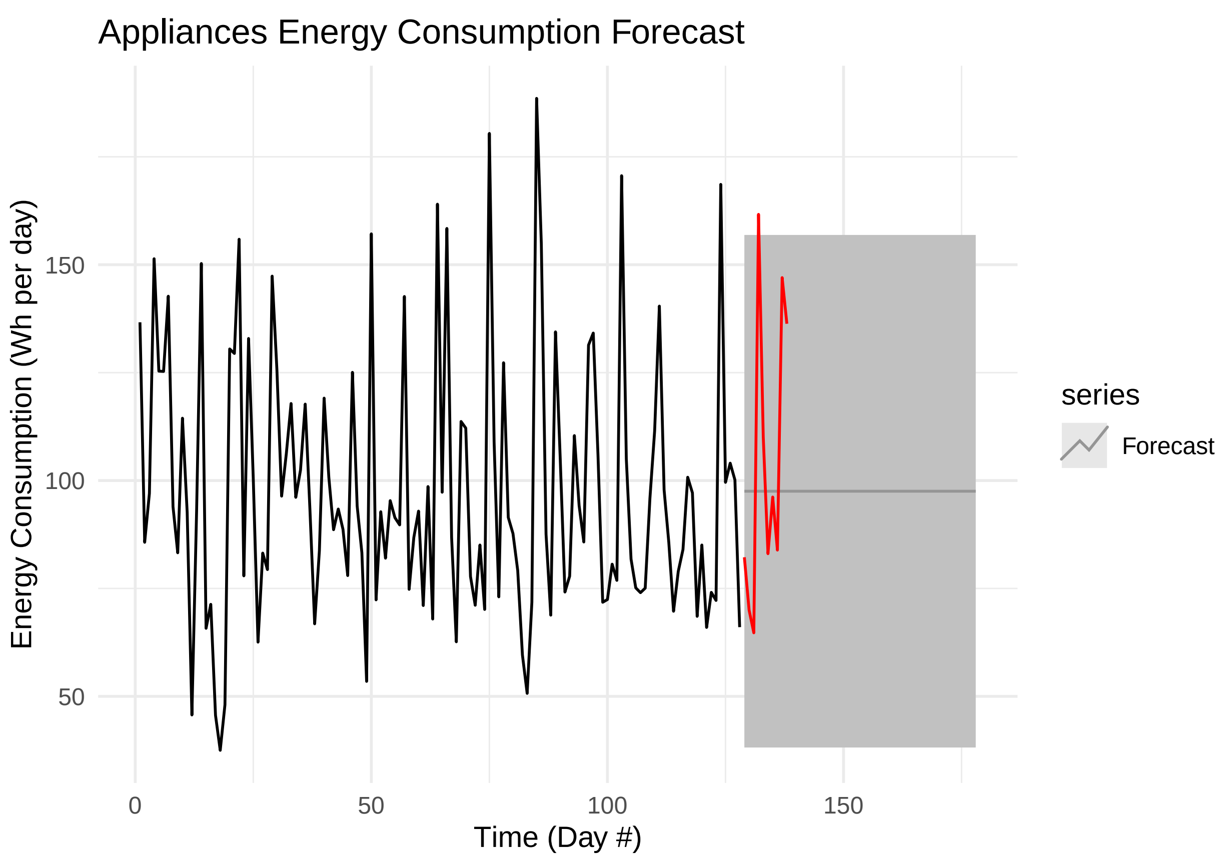
Algebraic Form of ARIMA(0,0,0) $(1 - B)^0 X_t = Z_t$ can be rewritten to $X_t = Z_t$

Yes, based on models we compared, ARIMA(0,0,0) is preferred for the lower AICc and BIC but ARIMA(1,0,0) AICc and BIC is low enough to be considered to be used.

Forecasting

```
#Forecast
h <- 50
```

```r
forecast_arima <- forecast(arima, h = h, level = c(95))

# Convert Appliances to a time series
appliances_ts <- ts(train_data$Appliances, frequency = 1)

test_ts <- ts(test_data$Appliances, start = length(train_data$Appliances) + 1, frequency = 1)


# Plot CI and test Data
autoplot(appliances_ts) +
  autolayer(forecast_arima, series = "Forecast", PI = TRUE) +
  autolayer(test_ts, series = "Test Data", color = "red") +
  ggtitle("Appliances Energy Consumption Forecast") +
  xlab("Time (Day #)") +
  ylab("Energy Consumption (Wh per day)") +
  theme_minimal() +
  scale_color_manual(values = c("black", "blue", "red"))
```



Here we can see the test data that was left out of data separation fits well within the confidence interval except 1 lag which is possible due to 95% CI.

Conclusion

Our goal was to use Time series analysis to model and forecast energy consumption, and predictive model to visualize a predictive model and forecast of energy usage. In addition to test with test data to see the accuracy of the model. Yes these goals were achieved, we successfully applied time series model, selected best ARIMA model, and generated forecast and confidence interval with test data. Algebraic Form of ARIMA(0,0,0) $(1 - B)^0 X_t = Z_t$ can be rewritten to $X_t = Z_t$ I was able to use R library and class lecture slides to complete this project.

References

R documentation: Autolayer Methods, forecast, mutation, acf and Pacf, AIC

Summary of ACF and PACF Patterns for (S)ARIMA-2 pages.pdf on modules on canvas were used for model and p and q calculation.

Final Project Guidelines W25 for structure of the report.

Candanedo, Luis. "Appliances Energy Prediction." UCI Machine Learning Repository, 2017, https://doi.org/10.24432/C5VC8G.

GeeksforGeeks. (2022, March 22). How to perform an augmented Dickey-Fuller test in R. https://www.geeksforgeeks.org/how-to-perform-an-augmented-dickey-fuller-test-in-r/

Appendix

```r
library(dplyr)
library(readr)
library(ggplot2)
library(forecast)
library(tseries)

#Reading DATA
energydata_complete <- read_csv("energydata_complete.csv")
energydata_complete <- energydata_complete %>%
  mutate(date = as.Date(date)) %>%
  group_by(date) %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE))

#Train and Test
train_size <- nrow(energydata_complete) - 10
train_data <- energydata_complete[1:train_size, ]
train_data <- train_data %>%
  mutate(lag_date = date - min(date))
test_data <- energydata_complete[(train_size + 1):nrow(energydata_complete), ]

#Validation
dim(train_data)
dim(test_data)
range(train_data$date)
range(test_data$date)


# Plot
ggplot(train_data, aes(x = lag_date, y = Appliances)) +
  geom_line(color = "blue") +
  ggtitle("Energy Consumption Over Time (Training Set)") +
```

```r
  xlab("Lag (Days from Start)") +
  ylab("Appliances Energy (Wh)") +
  theme_minimal()

#Stationary
ts_train <- ts(train_data$Appliances, frequency = 1)
#ADF
adf_test <- adf.test(ts_train)
print(adf_test)

# Plot ACF and PACF
par(mfrow = c(1,2))
Acf(ts_train, main = "ACF of Energy Consumption (Appliances)")
Pacf(ts_train, main = "PACF of Energy Consumption (Appliances)")


# ARIMA(0,0,0)
arima <- Arima(train_data$Appliances, order = c(0, 0, 0))
summary(arima)

#  AIC
aic <- AIC(arima)
k <- length(coef(arima))
n <- length(train_data$Appliances)
aicc <- aic + (2 * k * (k + 1)) / (n - k - 1)
aicc

# Residuals
checkresiduals(arima)

# ARIMA(1,0,0)
arima1 <- Arima(train_data$Appliances, order = c(1, 0, 0))
summary(arima1)

#  AIC
aic1 <- AIC(arima1)
k1 <- length(coef(arima1))
n1 <- length(train_data$Appliances)
aicc1 <- aic1 + (2 * k1 * (k1 + 1)) / (n1 - k1 - 1)
aicc1

# Residuals
checkresiduals(arima1)


#Forecast
h <- 50
forecast_arima <- forecast(arima, h = h, level = c(95))

# Convert Appliances to a time series
appliances_ts <- ts(train_data$Appliances, frequency = 1)

test_ts <- ts(test_data$Appliances, start = length(train_data$Appliances) + 1, frequency = 1)
```

```r
# Plot CI and test Data
autoplot(appliances_ts) +
  autolayer(forecast_arima, series = "Forecast", PI = TRUE) +
  autolayer(test_ts, series = "Test Data", color = "red") +
  ggtitle("Appliances Energy Consumption Forecast") +
  xlab("Lag (Day #)") +
  ylab("Energy Consumption (Wh per day)") +
  theme_minimal() +
  scale_color_manual(values = c("black", "blue", "red"))
```